

Internship Project Submission

“Privacy Notes”

Your Secure, Encrypted Note-Taking App



Submitted By:

Dhanraj Sahu

Web Development Intern

rajofficials23@gmail.com

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Privacy Notes: Your Secure, Encrypted Note-Taking App

1. Introduction

Welcome to Privacy Notes, a modern, privacy-first note-taking application designed to keep your personal thoughts and information truly private. In an age where data breaches and privacy concerns are rampant, Privacy Notes offers a robust solution by ensuring all your notes are encrypted directly on your device before they are ever stored. This means your data remains confidential, accessible only by you, and secure from external threats.

Whether you're jotting down quick ideas, detailed plans, or sensitive information, Privacy Notes provides a safe and reliable environment. It's built with an "offline-first" approach, guaranteeing access to your notes even without an internet connection, while offering optional cloud synchronization for backup and cross-device access with enhanced security.

2. Key Features at a Glance

- **Client-Side AES Encryption:** The cornerstone of Privacy Notes' security. All your notes are automatically encrypted using the Advanced Encryption Standard (AES) within your browser, *before* they are written to local storage or any optional cloud service. Your encryption key is managed locally and never leaves your device, putting you in complete control.
- **Offline-First Accessibility:** Your notes are stored securely using IndexedDB (or local Storage) directly in your browser. This design choice ensures that you can create, view, and edit your notes at anytime, anywhere, even if you don't have an active internet connection.
- **Intuitive Note Management:** A clean and straightforward user interface allows for seamless creation, reading, updating, and deleting of notes (CRUD operations).
- **Enhanced Organization Tools:**
 - **Powerful Search:** Quickly locate specific notes using keywords or phrases.
 - **Note Pinning:** Mark important notes to keep them easily accessible at the top of your list.
 - **Archiving Functionality:** Declutter your workspace by archiving older or less frequently used notes without deleting them permanently.
- **Optional Secure Cloud Synchronization:** For those who desire backup and seamless access across multiple devices, PrivacyNotes offers an optional integration with cloud services. This feature comes with optional authentication, adding an extra layer of security to your synchronized data.

PrivacyNotes is built using a modern and robust web technology stack, ensuring a responsive, secure, and scalable application.

- ## Client-Side Encryption Flow

```

graph LR
    Input[User Input Meeting notes: Q4 Goals, Budget Review] --> Step1[1. User Creates/Edits Note]
    Step1 --> App[PrivacyNotes Application Browser]
    subgraph App [PrivacyNotes Application Browser]
        direction TB
        Encrypt[2. Encrypts Data]
        subgraph CryptoJS [CryptoJS AES Encryption]
            Key[Local Encryption Key Stored Client-Side]
        end
        Ciphertext[Encrypted Data Ciphertext]
        Hex[U2FsdHkx1+...U2FsdHkx1+...]
        Encrypt --> Key
        Encrypt --> Ciphertext
        Ciphertext --> Hex
    end
    App --> IndexedDB[(IndexedDB or LocalStorage)]
    App --> CloudSync[Optional Cloud Sync Service]
    CloudSync --> Step4[4. Optional: Syncs Encrypted Data]
    Step4 --> Cloud[Cloud]
    Cloud --> DecryptionView[Decryption on View]
    DecryptionView --> Step5[5. Decrypts & Displays]
    Step5 --> Output[User View Meeting notes: Q4 Goals, Budget Review]
    
```